

Q1. A university stores the names of students in a string array and their corresponding marks in an integer array. Write a program that uses a method to calculate the average mark of all students. Write another method to count the number of vowels in a given student name. The program should display the names of all students who scored above the average, along with the number of vowels in their names. If necessary, a simple recursive method may be used to calculate the total sum of all marks.

Q2. A system maintains a list of usernames stored in a string array. A username is considered valid if it contains only letters and digits and has a minimum length of six characters. Write a program that uses one method to check whether a username contains only letters and digits and another method to determine the length of a username without using the built-in `length()` method. The program should display all valid usernames from the list. A basic recursive approach may optionally be used to count the number of digits in a username.

Q3. A weather monitoring center records the daily temperatures of a week in an integer array. Write a program that uses methods to determine the highest temperature, the lowest temperature, and the average temperature for the week. The program should then print all the days where the temperature was higher than the average. Finally, display a summary message converted to uppercase using a separate method.

Q4. A sentence is provided by the user as input and stored in a string. Write a program that uses a method to split the sentence into individual words and store them in a string array. Another method should be used to find the longest word in the sentence, and a separate method should count the total number of words. The program should display the longest word along with the total word count.

Q5. A shop stores the prices of its products in an integer array. Write a program that uses a method to calculate the total price of all products and another method to increase each product price by five percent, ensuring that no price exceeds 1000 after the increase. Additionally, use a method to determine the second highest price from the updated list. Finally, display the modified price array.

Q6. A library stores the titles of books in a string array and the number of times each book has been borrowed in an integer array. Write a program that uses a method to calculate the average number of borrows. Write another method to find the title with the

highest borrow count. The program should display all book titles that have been borrowed more than the average number of times along with their borrow counts.

Q7. An online survey collects feedback comments from users and stores them in a string array. Write a program that uses a method to count the number of words in a given comment and another method to check whether a comment contains a specific keyword provided by the user. The program should display all comments that contain the keyword along with their corresponding word counts.

Q8. A class teacher stores the attendance record of students for a month in an integer array, where each element represents the number of days a student was present. Write a program that uses a method to calculate the class average attendance. Write another method to determine whether a student is eligible to sit for the final exam, where eligibility requires attendance of at least 75 percent of the total classes. The program should display the attendance of all eligible students.